Stephanie Mui

CONTACT INFORMATION

Georgia Institute of Technology
Department of Mathematics
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RESEARCH INTERESTS

Convex geometry, Geometric analysis, Partial differential equations, and High-dimensional analysis

EDUCATION

Courant Institute of Mathematical Sciences, New York University

Ph.D. in Mathematics, May 2023

- Dissertation Topic: The *Lp* Aleksandrov problem and the *Lp* dual Minkowski problem for negative p
- Advisor: Deane Yang and Gaoyong Zhang

George Mason University

M.S. in Mathematics, May 2017

- Thesis Topic: Nash-Kuiper Surfaces
- Advisor: Sean Lawton

B.A. in Mathematics, May 2016

- Summa cum laude

AFFILIATIONS

Georgia Institute of Technology

Visiting Assistant Professor, 2026-present
NSF Postdoctoral Fellow, 2024 - 2026
Hale Visiting Assistant Professor, 2023 - 2024

Courant Institute of Mathematical Sciences, New York University

Ph.D. student and TA, 2017 - 2023

PUBLICATIONS

K. Boroczky, S. Mui, G. Zhang, *The weighted dual Minkowski problem under group symmetry*, arXiv:2609.20047, (2026). submitted.

S. Mui, L. Qin, S. Wang, *On the Minkowski problem with respect to a mixed Euclidean-Gaussian density*, arXiv:2607.27618, (2026). submitted.

M. Fernandez, G. Livshyts, S. Mui, *On the smallest singular value of log-concave random matrices*, arXiv:2508.17745, (2026). submitted.

K. Boroczky, A. Kovacs, S. Mui, G. Zhang, *Dual curvature density equation with group symmetry*, J. Diff. Eq., **465** (2026).

S. Mui, *On the L^p dual Minkowski problem for $-1 < p < 0$,* Calc. Var. Par. Diff. Eq., **64** (2023).

S. Mui, *On the L^p Aleksandrov problem for negative p,* Adv. Math. **408** (2022).

Talks

AMS Special Session on Interface Between Calculus of Variations, Harmonic Analysis, and Complex Analysis, Kennesaw, Georgia (October 2026)

The log-Brunn-Minkowski conjecture: State of the art and ramifications, Erdos center, Budapest (July 2026)

"Brunn-Minkowski Theories: 80th birthday of Erwin Lutwak", Shanghai University, Shanghai, China (June 2026)

Convexity Seminar, NYU Courant Institute, New York, New York (April 2026)

Convex Geometry Seminar, Hunan University, Changsha, China. (December 2025)

Convex Geometric Analysis and Stochastic Geometry, Tianyuan Mathematics Research Center, Kunming, Yunnan. (November 2025)

AMS Special Session on Convexity, Probability, and Analysis. St. Louis, Missouri. (October 2025)

Mathematical Congress of Special Session on Convexity, High Dimensional Probability and Applications. Miami, Florida. (July 2025)

Convex Geometry Seminar, TU Wien. Vienna, Austria. (December 2024)

Harmonic Analysis and Convexity, ICERM. Providence, Rhode Island. (December 2024)

International Conference on Convex Geometry and Related topics. Xi'an, Shaanxi. (November 2024)

AMS Special Session Probabilistic and Analytic Aspects in Convexity. Albany, New York. (October 2024)

Conference on Convex Geometry and Related PDEs. Changsha, Hunan. (May 2024)

Georgia Tech Analysis Workshop, Georgia Tech (December 2023)

Seminar der Arbeitsgruppen Diskrete Mathematik/ Geometrie & Diskrete Geometrie, TU Berlin, (July 2023)

Convex Geometry - Analytic Aspects, Istituto Nazionale di Alta Matematica (INdAM). (June 2023)

Yale Geometric Analysis and Applications Seminar, Yale University. (April 2023)

Syracuse Analysis Seminar, Syracuse University. (April 2023)

AMS Southeastern Sectional Meeting: Special Session on High-dimensional Convexity and Probability, Georgia Tech. (March 2023)

Harmonic Analysis Methods in Geometric Tomography, ICERM. (September 2022)

Canadian Math Society Summer Meeting: Session on Convex geometry and Partial Differential Equations, Memorial University of Newfoundland. (June 2022)

Workshop in Convexity and High-dimensional probability, Georgia Tech, Short talk. (May 2022)

Online Asymptotic Geometric Analysis Seminar. (March 2022)

Convex Geometry and its Applications, Oberwolfach Research Institute for Mathematics (MFO). (December 2021)

Geometry Labs United Conference, ICERM, Lightning talk and poster. (July 2020)

Undergraduate Mathematics Symposium, University of Illinois at Chicago. (October 2016)

Teaching Experience

Spring	2024	Instructor (Georgia Tech), Probability Theory
Fall	2023	Instructor (Georgia Tech), Finite Mathematics
Fall	2022	Teaching Assistant (NYU), Analysis
Spring	2022	Teaching Assistant (NYU), Honors Analysis I and II
Fall	2021	Teaching Assistant (NYU), Analysis
Spring	2021	Teaching Assistant (NYU), Analysis
Spring	2017	Teaching Assistant (GMU), Calculus I

Service

2024–2026	Supporting Faculty for the Atlanta Undergraduate Research Mathematics Seminar
2023–2025	Mentor at Directed Reading Program for Georgia Tech Undergrads for Kalev Martinson, Sahil Vora, and Anand Singh
2020–2021	Mentor for NYU Mathematics Master's Students for Aadya Bhatnagar and Zachary Brogie
2015–2017	Research Intern at the Mason Experimental Geometry Lab Volunteered for outreach activities hosted by lab

Honors and Awards

2024–2026	NSF Mathematical Sciences Postdoctoral Research Fellowship National Science Foundation (NSF)
2023	Bella Manel Award Courant Institute of Mathematical Sciences (NYU)
2021	US Junior Oberwolfach Fellow Oberwolfach Research Institute for Mathematics (MFO)
2018–2020	Summer Opportunity Fellowship New York University Graduate School of Arts and Sciences
2017–2022	Henry MacCracken Fellowship New York University Graduate School of Arts and Sciences
2016	First Place AMS Menger Award Intel International Science and Engineering Fair